

Problem 9.8.32

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Question

Question: The centre of the circle passing through the point $(0, 1)$ and touching the curve $y = x^2$ at $(2, 4)$ is

- (a) $\left(\frac{-16}{5}, \frac{27}{10}\right)$
- (b) $\left(\frac{-16}{7}, \frac{53}{10}\right)$
- (c) $\left(\frac{-16}{5}, \frac{53}{10}\right)$
- (d) none of these

Input Variables:

Parameters of the Circle:

Where $\mathbf{c}$ is the centre of circle.

$$\mathbf{V}_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \mathbf{u}_1 = -\mathbf{c} \quad (2.1)$$

Parameters of the Parabola:

$$\mathbf{V}_2 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \mathbf{u}_2 = \begin{pmatrix} 0 \\ -\frac{1}{2} \end{pmatrix}, f_2 = 0 \quad (2.2)$$

$$\mathbf{P} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \mathbf{Q} = \begin{pmatrix} 2 \\ 4 \end{pmatrix} \quad (2.3)$$

Solution

Step 1: Find one equation for $\mathbf{c}$

Points $\mathbf{P}$ and $\mathbf{Q}$ lie on the circle so,

$$\mathbf{P}^\top \mathbf{V}_1 \mathbf{P} - 2\mathbf{c}^\top \mathbf{P} + f_1 = 0, \quad (2.4)$$

$$\mathbf{Q}^\top \mathbf{V}_1 \mathbf{Q} - 2\mathbf{c}^\top \mathbf{Q} + f_1 = 0, \quad (2.5)$$

Subtracting (0.4)-(0.5),

$$(\mathbf{P}^\top \mathbf{V}_1 \mathbf{P} - \mathbf{Q}^\top \mathbf{V}_1 \mathbf{Q}) - 2\mathbf{c}^\top (\mathbf{P} - \mathbf{Q}) = 0, \quad (2.6)$$

$$2\mathbf{c}^\top (\mathbf{P} - \mathbf{Q}) = \mathbf{P}^\top \mathbf{V}_1 \mathbf{P} - \mathbf{Q}^\top \mathbf{V}_1 \mathbf{Q}. \quad (2.7)$$

$$-2\mathbf{c}^\top \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} - \begin{pmatrix} 2 & 4 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 4 \end{pmatrix}. \quad (2.8)$$

$$\begin{pmatrix} 4 & 6 \end{pmatrix} \mathbf{c} = 19 \quad (2.9)$$

Step 2: Find the equation of tangent through (2,4) at Parabola

$$(\mathbf{V}_2 \mathbf{Q} + \mathbf{u}_2)^\top \mathbf{x} + \mathbf{u}_2^\top \mathbf{Q} + f_2 = 0, \quad (2.10)$$

$$\left(\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 2 \\ 4 \end{pmatrix} + \begin{pmatrix} 0 \\ -\frac{1}{2} \end{pmatrix} \right)^\top \mathbf{x} + \begin{pmatrix} 0 \\ -\frac{1}{2} \end{pmatrix}^\top \begin{pmatrix} 2 \\ 4 \end{pmatrix} + 0 = 0, \quad (2.11)$$

$$\left(2 \quad -\frac{1}{2} \right) \mathbf{x} = 2. \quad (2.12)$$

Solution

Step 3: Find the equation of normal to circle at (2,4)

Let the equation of normal to circle at (2,4) be:

$$\mathbf{n}^T \mathbf{x} = 1 \quad (2.13)$$

The normal is perpendicular to the previously calculated tangent in (0.12)

$$\mathbf{n}^T \begin{pmatrix} 2 \\ -\frac{1}{2} \end{pmatrix} = 0 \quad (2.14)$$

It passes through (2,4)

$$\mathbf{n}^T \begin{pmatrix} 2 \\ 4 \end{pmatrix} = 1 \quad (2.15)$$

Row Reduction Steps:

$$\left(\begin{array}{cc|c} 2 & -\frac{1}{2} & 0 \\ 2 & 4 & 1 \end{array} \right) \quad (2.16)$$

Solution

Perform the row operation $R_2 \rightarrow R_2 - R_1$:

$$\left(\begin{array}{cc|c} 2 & -\frac{1}{2} & 0 \\ 0 & \frac{9}{2} & 1 \end{array} \right) \quad (2.17)$$

Perform the row operation $R_1 \rightarrow R_1 + \frac{R_2}{9}$:

$$\left(\begin{array}{cc|c} 2 & 0 & \frac{1}{9} \\ 0 & \frac{9}{2} & 1 \end{array} \right) \quad (2.18)$$

$$\mathbf{n} = \frac{1}{18} \begin{pmatrix} 1 \\ 4 \end{pmatrix} \quad (2.19)$$

Equation of normal is :

$$(1 \ 4) \mathbf{x} = 18 \quad (2.20)$$

Solution

Step 4: Find $\mathbf{c}$

Centre lies on the normal (2.20)

$$\begin{pmatrix} 1 & 4 \end{pmatrix} \mathbf{c} = 18 \quad (2.21)$$

from equation (2.9)

$$\begin{pmatrix} 4 & 6 \end{pmatrix} \mathbf{c} = 19 \quad (2.22)$$

Row reduction steps:

$$\left(\begin{array}{cc|c} 1 & 4 & 18 \\ 4 & 6 & 19 \end{array} \right) \quad (2.23)$$

Perform the row operation $R_2 \rightarrow R_2 - 4R_1$:

$$\left(\begin{array}{cc|c} 1 & 4 & 18 \\ 0 & -10 & -53 \end{array} \right) \quad (2.24)$$

Solution

Perform the row operation $R_1 \rightarrow R_1 + \frac{2}{5}R_2$:

$$\left(\begin{array}{cc|c} 1 & 0 & -\frac{16}{5} \\ 0 & -10 & -53 \end{array} \right) \quad (2.25)$$

$$\mathbf{c} = \left(\begin{array}{c} -\frac{16}{5} \\ \frac{53}{10} \end{array} \right) \quad (2.26)$$

Final Answer:

The centre of the circle is $\left(\begin{array}{c} -\frac{16}{5} \\ \frac{53}{10} \end{array} \right)$.

Graph

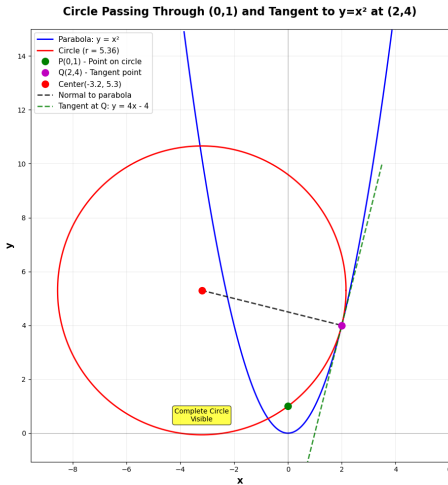


Figure: Vector Representation

```
#include "matfun.h"
#include <math.h>

// Solve 2x2 system Ax = b using simple method
int solve_2x2(double A[4], double b[2], double x[2]) {
    // A = [a11, a12, a21, a22] (row-major)
    double det = A[0]*A[3] - A[1]*A[2];

    if (fabs(det) < 1e-10) return 0; // No solution

    x[0] = (b[0]*A[3] - b[1]*A[1]) / det;
    x[1] = (A[0]*b[1] - A[2]*b[0]) / det;

    return 1; // Success
}
```

```
// Calculate distance between two points
double distance(double x1, double y1, double x2, double y2) {
    double dx = x1 - x2;
    double dy = y1 - y2;
    return sqrt(dx*dx + dy*dy);
}
```

```
// Solve the circle-parabola problem
int solve_problem(double center[2], double *radius) {
    // System:  $4cx + 6cy = 19$ ,  $cx + 4cy = 18$ 
    double A[4] = {4.0, 6.0, 1.0, 4.0};
    double b[2] = {19.0, 18.0};

    if (!solve_2x2(A, b, center)) return 0;

    // Calculate radius from center to (0,1)
    *radius = distance(center[0], center[1], 0.0, 1.0);

    return 1;
}
```

```
import ctypes
import numpy as np
import matplotlib.pyplot as plt

# Load C library
lib = ctypes.CDLL('./libmatfun.so')

# Set up function signatures
lib.solve_problem.argtypes = [ctypes.POINTER(ctypes.c_double),
                               ctypes.POINTER(ctypes.c_double)]
lib.solve_problem.restype = ctypes.c_int
```

```
# Main execution
if __name__ == "__main__":
    # Call C function to solve the problem
    center = (ctypes.c_double * 2)()
    radius = ctypes.c_double()

    result = lib.solve_problem(center, ctypes.byref(radius))

    if result:
        cx, cy, r = center[0], center[1], radius.value

        print(f"Center: ({cx:.6f}, {cy:.6f})")
        print(f"Radius: {r:.6f}")

    # Simple plotting
    fig, ax = plt.subplots(figsize=(10, 8))
```

```
# Plot parabola  $y = x$ 
x_para = np.linspace(-8, 4, 500)
y_para = x_para**2
mask = (y_para <= 12) # Limit y range
ax.plot(x_para[mask], y_para[mask], 'b-', linewidth=2,
        label='y = x')

# Plot circle
theta = np.linspace(0, 2*np.pi, 500)
x_circle = cx + r * np.cos(theta)
y_circle = cy + r * np.sin(theta)
ax.plot(x_circle, y_circle, 'r-', linewidth=2, label='
    Circle')

ax.plot(0, 1, 'go', markersize=8, label='P(0,1)')
ax.plot(2, 4, 'mo', markersize=8, label='Q(2,4)')
ax.plot(cx, cy, 'ro', markersize=8, label=f'Center({cx:.1
    f},{cy:.1f})')
```



```
# Normal line
ax.plot([cx, 2], [cy, 4], 'k--', alpha=0.7, label='Normal
    ')

# Tangent line  $y = 4x - 4$ 
x_tan = np.linspace(0.5, 3.5, 50)
y_tan = 4*x_tan - 4
ax.plot(x_tan, y_tan, 'g--', alpha=0.7, label='Tangent')

# Set limits and format
ax.set_xlim(cx-r-1, cx+r+1)
ax.set_ylim(cy-r-1, cy+r+1)
ax.grid(True, alpha=0.3)
ax.set_xlabel('x')
ax.set_ylabel('y')
ax.set_title('Circle-Parabola Problem (C Library)')
ax.legend()
ax.set_aspect('equal')
```

Python Code

```
import numpy as np
import matplotlib.pyplot as plt

def solve_circle_parabola_problem():
    """Find the center of the circle passing through (0,1) and
        touching  $y = x$  at (2,4)"""

    # Given points
    P = np.array([0, 1])
    Q = np.array([2, 4])

    # Set up system of equations
    # Constraint 1: Equal distances from center to P and Q
    #  $4cx + 6cy = 19$ 
    # Constraint 2: Center-to-Q line is normal to parabola at Q
    #  $cx + 4cy = 18$ 
```

```
A = np.array([[4, 6], [1, 4]])
b = np.array([19, 18])

# Solve for center
center = np.linalg.solve(A, b)
radius = np.linalg.norm(P - center)

return center, radius

def plot_solution():
    """Plot the parabola, circle, and key points with complete
       circle visible"""
    center, radius = solve_circle_parabola_problem()

    fig, ax = plt.subplots(1, 1, figsize=(12, 10))
```

Python Code

```
# Calculate proper plot limits to show entire circle
x_min = center[0] - radius - 1
x_max = max(center[0] + radius + 1, 6) # Ensure parabola
    portion is visible
y_min = center[1] - radius - 1
y_max = max(center[1] + radius + 1, 15) # Show some parabola

# Plot parabola y = x (extended range)
x_parabola = np.linspace(x_min, x_max, 1000)
y_parabola = x_parabola**2
# Only plot parabola where y is in our range
mask = (y_parabola >= y_min) & (y_parabola <= y_max)
ax.plot(x_parabola[mask], y_parabola[mask], 'b-', linewidth
        =2, label='Parabola: y = x')

# Plot complete circle
theta = np.linspace(0, 2*np.pi, 1000)
x_circle = center[0] + radius * np.cos(theta)
y_circle = center[1] + radius * np.sin(theta)
```

```
ax.plot(x_circle, y_circle, 'r-', linewidth=2, label=f'Circle  
(r = {radius:.2f})')

# Mark important points
ax.plot(0, 1, 'go', markersize=10, label='P(0,1) - Point on  
circle', zorder=5)
ax.plot(2, 4, 'mo', markersize=10, label='Q(2,4) - Tangent  
point', zorder=5)
ax.plot(center[0], center[1], 'ro', markersize=10,  
        label=f'Center({center[0]:.1f}, {center[1]:.1f})',  
        zorder=5)

# Draw normal line from center to Q
ax.plot([center[0], 2], [center[1], 4], 'k--', linewidth=2,  
        alpha=0.8, label='Normal to parabola')

# Draw tangent line at Q ( $y = 4x - 4$ )
x_tangent = np.linspace(0.5, 3.5, 100)
y_tangent = 4 * x_tangent - 4
```

```
ax.plot(x_tangent, y_tangent, 'g--', linewidth=2,
        alpha=0.8, label='Tangent at Q:  $y = 4x - 4$ ')

# Set plot limits to show complete circle
ax.set_xlim(x_min, x_max)
ax.set_ylim(y_min, y_max)

# Enhance plot appearance
ax.grid(True, alpha=0.4, linestyle='-', linewidth=0.5)
ax.set_xlabel('x', fontsize=14, fontweight='bold')
ax.set_ylabel('y', fontsize=14, fontweight='bold')
ax.set_title('Circle Passing Through (0,1) and Tangent to  $y=x$ 
            at (2,4)',
            fontsize=16, fontweight='bold', pad=20)
```

```
# Add axes lines through origin
ax.axhline(y=0, color='black', linewidth=0.8, alpha=0.3)
ax.axvline(x=0, color='black', linewidth=0.8, alpha=0.3)

# Legend with better positioning
ax.legend(loc='upper left', fontsize=11, framealpha=0.9)

# Make axes equal to avoid distortion
ax.set_aspect('equal', adjustable='box')

# Add some annotations
ax.annotate('Complete Circle\nVisible',
            xy=(center[0], center[1]-radius+0.5),
            fontsize=10, ha='center',
            bbox=dict(boxstyle="round,pad=0.3", facecolor="
                    yellow", alpha=0.7))
```

```
plt.tight_layout()
plt.show()

return fig, ax

if __name__ == "__main__":
    center, radius = solve_circle_parabola_problem()

    print(f"Center: ({center[0]:.6f}, {center[1]:.6f})")
    print(f"Center (exact): (-16/5, 53/10)")
    print(f"Radius: {radius:.6f}")

    plot_solution()
```